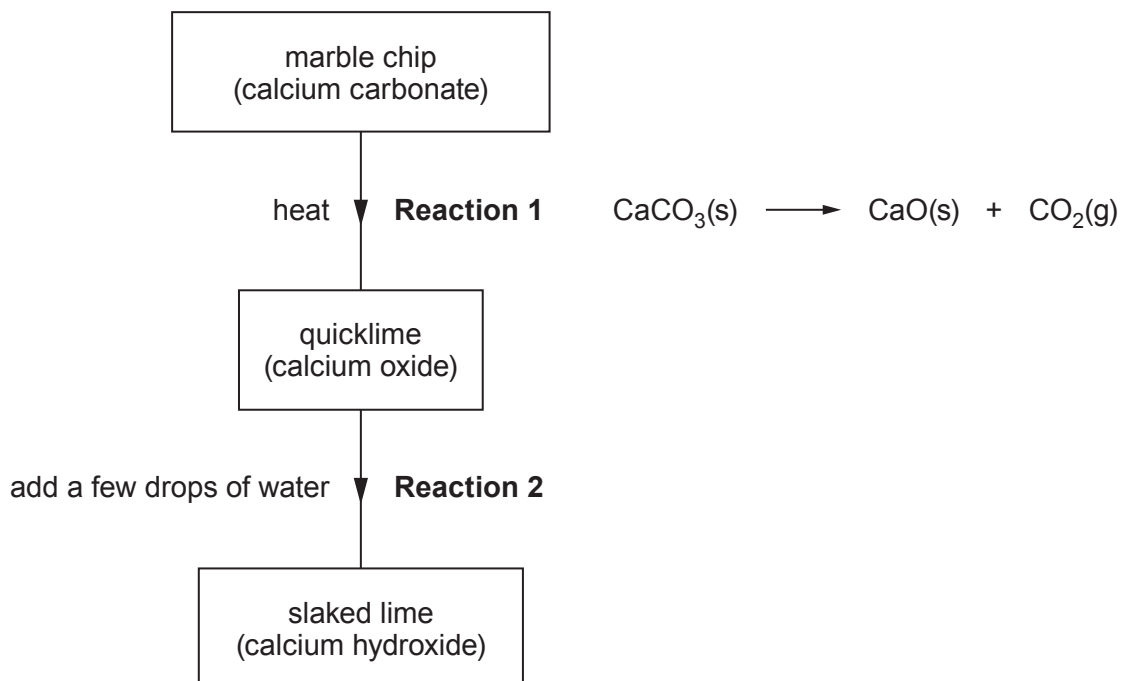


4. (a) The flow diagram shows the reactions necessary to make slaked lime (calcium hydroxide) from a marble chip (calcium carbonate).



- (i) In **Reaction 1**, one compound breaks down when heated to form two products.

Complete the name of this type of reaction.

[1]

thermal

- (ii) Give **two** observations you would expect to make as **Reaction 2** happens.

[2]

1

2

- (iii) Complete the symbol equation for the reaction between calcium oxide and water in **Reaction 2**.

[1]

..... + H_2O $\longrightarrow$



Examiner
only

(b) (i) The relative formula mass of calcium hydroxide is 74.

Calculate the number of moles of calcium hydroxide in 2.96 g. [2]

Number of moles = mol

(ii) Explain why calcium hydroxide is used to treat some soils. [2]

.....

.....

.....

3410UA01
11

8

